Sparse-Feature Instability in Direct MLP Penetration Prediction

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Message

Claim: if sparse operating variables such as nozzle diameter and injection pressure are fed directly into a smooth MLP, the network can invent continuous interpolation behavior that is not supported by the data and is not physically credible.

How we will show this:

- first show the support topology: where the data manifold is dense, and where it has holes
- then compare dense MLP sweeps against secants implied by only the observed sparse levels
- finally show that the instability persists over time, especially for nozzle diameter

Why This Is a Sparse-Feature Problem

Observed support from the clean audit table:

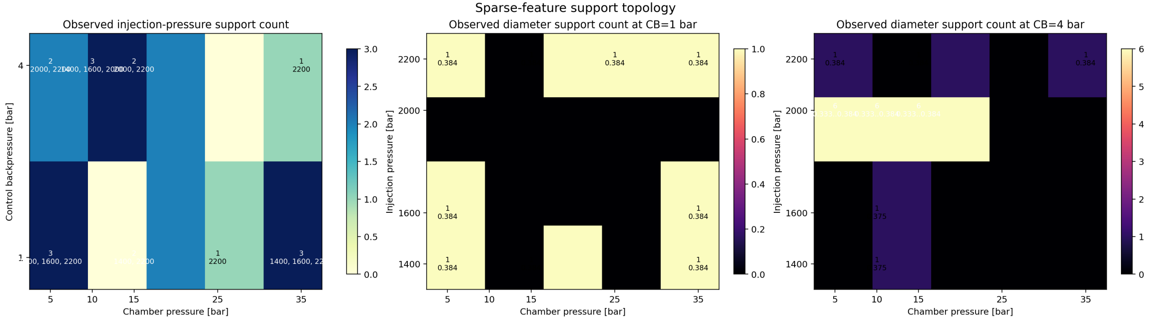
- injection pressure has only **4** levels: 1400, 1600, 2000, 2200 bar
- nozzle diameter has only **6** levels: 0.333, 0.348, 0.355, 0.365, 0.375, 0.384 mm
- diameter is *not* available uniformly across pressure combinations
- all six diameter levels only appear on the narrow family

$$(P_{\text{inj}} = 2000,\ \ C_b = 4,\ \ P_{\text{ch}} \in \{5, 10, 15\})$$

Implication:

- a dense gradient with respect to diameter is mostly an interpolation artifact between a few discrete nozzle levels
- even injection-pressure sensitivity is only trustworthy on a small number of supported slices

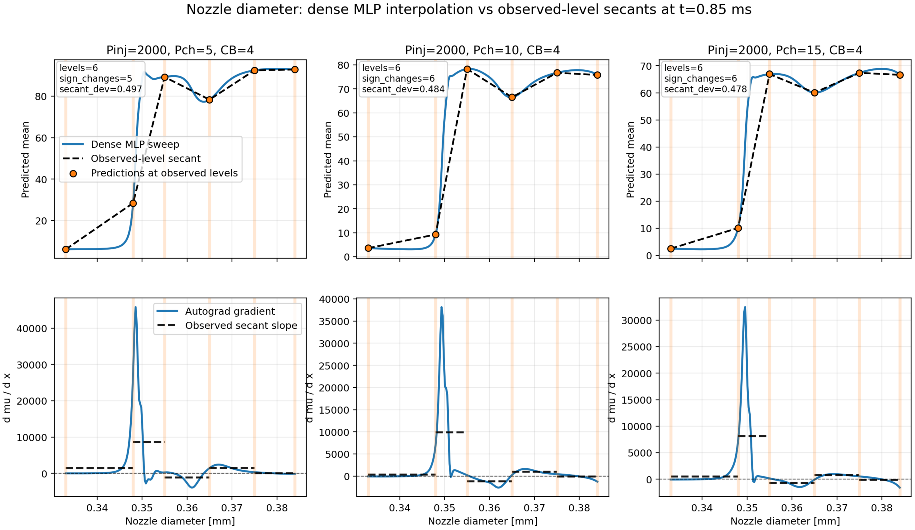
Support Topology of Sparse Features



Interpretation of the Support Topology

- left panel: injection pressure support count as a function of chamber pressure and control backpressure
- right panels: diameter support count as a function of injection pressure and chamber pressure, split by control backpressure
- for most pressure triplets, diameter count is $\mathbf{1}$; the model sees one nozzle level only
- therefore a derivative like $\partial\mu/\partial d$ cannot be strongly constrained by data almost anywhere
- this is the structural reason the MLP is free to create arbitrary smooth curves between nozzle levels

Nozzle Diameter: Dense Interpolation vs. Observed Secants



Why Diameter Is the Strongest Counterexample

Slide reading guide:

- blue curve: dense MLP sweep over diameter
- orange markers: MLP predictions only at actually observed diameter levels
- black dashed line: piecewise secant implied by those observed levels only

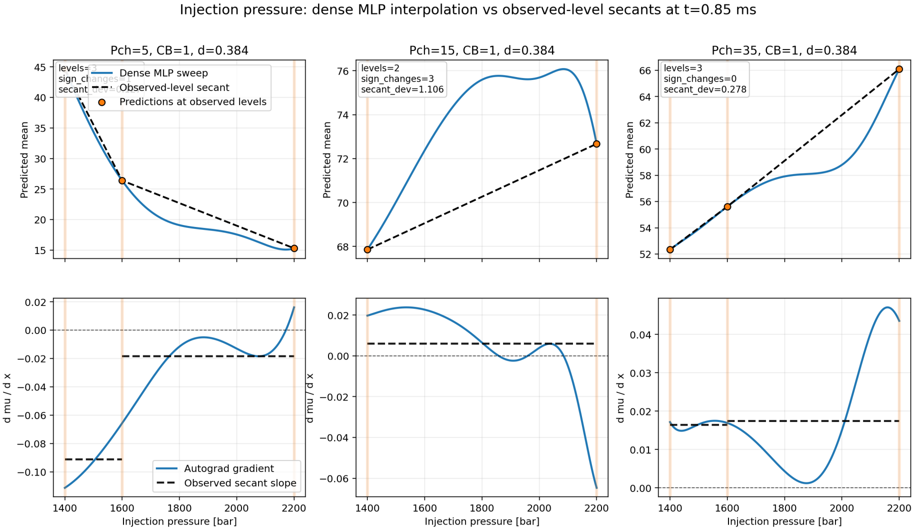
Observed behavior at $t = 0.85$ ms:

- gradient sign changes $=$ **5–6** across the three supported diameter slices
- worst secant-deviation ratio is about **0.48–0.50**
- $p95\ |d^2\mu/dd^2|$ reaches order 10^7

Interpretation:

- the dense MLP curve bends strongly between sparse nozzle levels
- this curvature is not implied by the observed level-to-level trend
- physically, this looks like invented interpolation rather than a learned monotonic nozzle-size law

Injection Pressure: Better Than Diameter, Still Not Safe



Injection Pressure Still Shows Unsupported Behavior

Observed behavior at $t = 0.85$ ms:

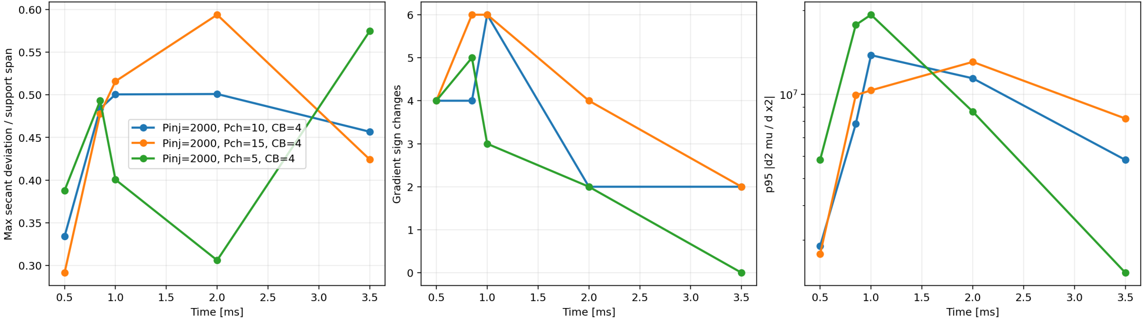
- the slice `Pch=15, CB=1, d=0.384` has only **2** pressure levels
- on that slice, worst secant-deviation ratio is > 1.0
- some slices still show **1–3** gradient sign changes, despite expected positive sensitivity to P_{inj}

Interpretation:

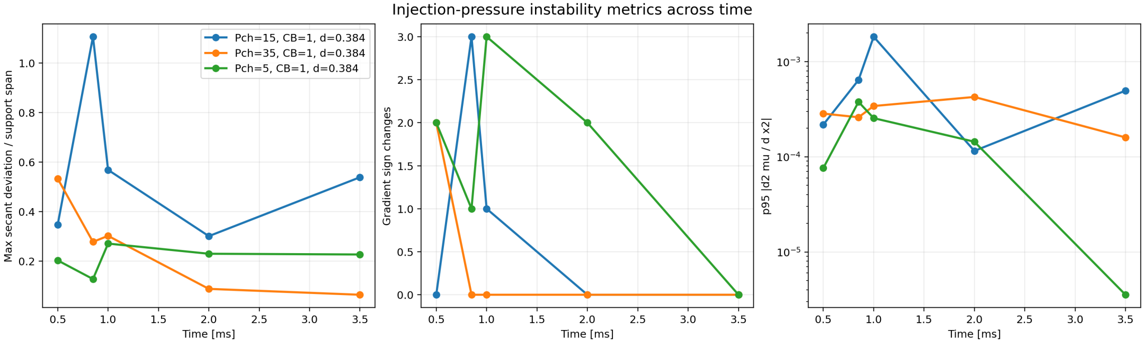
- injection pressure is less pathological than diameter, but the direct-feature MLP still invents smooth structure between sparse pressure levels
- on weakly supported slices, the continuous derivative field becomes a modeling choice, not a data-driven conclusion

Instability Over Time: Nozzle Diameter

Nozzle diameter instability metrics across time



Instability Over Time: Injection Pressure



Time-Series Interpretation

- diameter instability is persistent, not a single-time anomaly
- across time, diameter keeps large secant-deviation ratios, many sign changes, and very large curvature
- injection pressure is more slice-dependent: some families remain relatively smooth, while others become highly unreliable when support is thin
- this means the issue is not just local optimization noise; it is a systematic consequence of sparse-feature geometry in the training set

Engineering Takeaway

What these slides support:

- direct continuous learning on sparse nozzle diameter is not trustworthy for physical sensitivity analysis
- direct continuous learning on sparse injection pressure can also fail on weakly supported slices
- an MLP can fit training loss while still producing anti-physical derivative structure between sparse feature levels

Practical next steps:

- treat diameter as a discrete nozzle identity or hierarchical effect, not a free continuous variable
- constrain sensitivity using monotonic priors / secant regularization / family-wise models
- always pair dense sweeps with support maps before interpreting gradients physically